

List of Corrections

Submitted in response to the Degree Committee decision of 11 May 2026 (minor corrections).

Corrections are listed in order of appearance in the thesis. Original page numbers refer to the thesis version examined at viva (22 April 2026); new page numbers refer to the corrected, resubmitted thesis.

Chapter 1, p.6 → Chapter 1, pp.6–7

Corrected an overstatement of Expected Utility Theory's standing in modern decision-making research. The passage now describes EUT as seminal to the field and influential in shaping experimental paradigms, but superseded in contemporary behavioural economics and judgment-and-decision-making research by Cumulative Prospect Theory, with a new closing sentence citing Tversky & Kahneman (1992). A further new paragraph introduces Cumulative Prospect Theory's two key mechanisms — subjective probability weighting and loss aversion/reference-point dependence — addressing a separate comment that the introduction lacked discussion of Prospect Theory and loss aversion.

Chapter 1, p.8 → Chapter 1, p.9

Corrected a mischaracterisation of the Rescorla-Wagner model as having cause-effect, predictive structure. The model is now described as associative rather than causal, explicitly noted as not predictive of future states, and trial-level only. The following page's discussion of TD models notes extension beyond the trial as the unit of time, further reinforces the corrected characterisation.

Chapter 1, p.12 → Chapter 1, p.13

Added explicit "Real data" / "Simulated data" labels directly on the figure to visually distinguish panel A (real) from panels B–C (simulated).

— (new) → Chapter 1, pp.14–15

Citation to Sharpe's 2017 paper on sufficiency/necessity of DA for associative learning. Below, added a new paragraph on the interpretation of the computational role of dopaminergic activity: notes this thesis's focus on utility encoding, while naming other theoretical accounts (reward prediction error/RL, causal learning).

Chapter 2, p.35 → Chapter 2, p.37

Converted three long bullet points into equivalent prose, linked with "First," "Second," "Finally."

— (new) → Chapter 3, p.46

Added new Section 3.1.4, "Data and Code Availability", detailing where to obtain the processed data and analysis code. Section 3.1.5, "Animals", gained a new paragraph adapted from Hill et al. 2024 naming the Home Office licence and the full UK/Cambridge ethical oversight framework.

Chapter 3, p.46 → Chapter 3, p.48

Converted five long bullet points into equivalent prose.

Chapter 3, p.51 → Chapter 3, p.53

Added sentences to the main body (rather than the Figure 3.4 caption, to avoid bloating it) stating the criterion for ending a session: sessions continued until 5 consecutive errors on days meeting the 200-trial target, or for a further 15 minutes after the monkey's last response on shorter days, to prevent the animal learning that withholding participation would end a session early.

Chapter 3, p.63 → Chapter 3, p.65

Rebuilt Figure 3.10 to address unreadable, repeated axis labels. Y-axis labels enlarged and now shown only on the top-left panel, with a caption note explaining the omission; a secondary axis added labelling the right-hand strip as "starting bid decile".

Chapter 3, p.65 → Chapter 3, p.68

Rebuilt Figure 3.12. Row and column strip labels repositioned (rows to the left, columns to the bottom); axis tick numbers removed entirely, leaving only variable names labelling each row/column of the correlation matrix. Fixed a stale panel-letter reference in the caption left over from an earlier A/B split by monkey.

Chapter 3, p.71 → Chapter 3, p.74

Corrected Monkey V's single-fractal context effect ($d=0.4666$) from "small" to "medium", noting it rounds to Cohen's 0.5 medium threshold. Also expanded the surrounding discussion to consider relative valuation as the explanation, rather than attributing the difference solely to temporal value drift. Notes Monkey U shows no such effect, consistent with range adaptation typically being partial rather than complete even in canonical demonstrations. Link forward to section 4.3.2.

Chapter 3, p.76 → Chapter 3, p.78

Added paragraphs discussing the dissociation between monkeys in the effect of total liquid received during single-fractal sessions: Monkey V shows the expected positive effect, while Monkey U shows a non-significant reduction across the full session-level regression (though significantly positive in 9 of 22 individual sessions) for reasons that remain unclear. Notes this does not undermine the thesis's central claims, and that the two animals are not expected to replicate one another perfectly. Figure 3.17 also added correctly.

Chapter 3, p.87 → Chapter 3, pp.90, 92–93

Clarified the "tie-fighter" markers (point estimate $\pm$ error bars for the bundle-choice indifference point) against the adjacent BDM boxplots, rewriting the caption to make explicit. Added second figure with an

extended y-axis. Interpretive sentences explaining the 50% logistic-crossing definition and the indifference point as an independent WTP measure were moved from the caption into the body text to avoid caption overrun.

Chapter 3, p.88 → Chapter 3, p.91

Corrected Table 3.10's caption, which implied Figure 1.1 was the data source for the Bundle WTP indifference point. Caption rewritten to correctly cite Figures 1.1 and 2.2 only as conceptual illustrations of what an indifference point is.

Chapter 4, p.103 → Chapter 4, p.107

Corrected Figure 4.4's caption: fixed a backwards left/right (Monkey U/Monkey V) description; replaced the claim that y-axis scales differ, when none were ever shown, with an explanation that spike counts are normalised to each panel's own maximum, giving a common 0–1 scale.

Chapter 4, p.104 → Chapter 4, pp.106 & 108

Clarified why fewer neurons appear in the raster than the population-average plot of the unpredicted-reward response. Added the explicit inclusion criterion (>5 free-reward trials, lenient Wilcoxon $p < 0.2$) and prose explaining that omitted cells were under-tested to avoid satiation, not classed as non-dopaminergic or non-responsive — they may well have been free-reward responsive had they been tested sufficiently.

Chapter 4, p.139 → Chapter 4, p.119

Added a short paragraph justifying the use of binned bids in this analysis: averaging within a bin reduces the influence of trial-to-trial motor/reporting noise, makes the analysis comparable to the inherently discrete binary-choice paradigms, and enables clear visualisation with an interpretable standard error at each point. Leads directly into the mixed-effects model, which uses the continuous, unbinned bid as the primary statistical test (Equation 4.6). LME is then used throughout this chapter, and added mixed-effects results confirming the central bid-dopamine claim once cell-level clustering is accounted for.

— (new) → Chapter 4, p.121

Added short paragraph after central results of bid-dopamine activity foreshadowing splitting by fractal level and comparing bids of similar magnitude.

— (new) → Chapter 4, p.123

Added a spaghetti plot (Figure 4.14) of per-cell mixed-effects predictions against the population fixed-effect, visualising the central bid-dopamine relationship at the trial level without binning.

— (new) → Chapter 4, p.126

Added brief discussion of the 'inverted U' shape of the Figure 4.17A visuals. Data investigation did not hint at anything more interesting than artefacts.

Chapter 4, p.124 → Chapter 4, p.131

Fixed a copy-paste error in Table 4.5 (formerly Table 4.3): intercept and slope values had been duplicated across all three reward-level rows per monkey rather than computed separately per level. Correct, distinct per-level values now shown; the corresponding values already reported in the main text were correct and unaffected.

— (new) → Chapter 4, p.127

Added a mixed-effects test for whether the dopamine response to the medium fractal differs between single-fractal and multi-fractal sessions; no significant effect of session context found for either monkey.

— (new) → Chapter 4, p.129

Added a mixed-effects regression confirming the dopamine response tracks offer value within single-fractal sessions: significant for Monkey V, a positive but non-significant trend for Monkey U. Added synthesis noting dopamine tracks value even with no other reward magnitude present to provide context, unlike the monkeys' bids, which do show a context effect.

— (new) → Chapter 4, p.132

Added mixed-effects results (Table 4.6) for the dopamine-bid relationship split by offer value. Significant for five of six monkey-by-fractal combinations; the exception, Monkey V's high-value fractal, was the naive regression's strongest result but is no longer significant once cell-level clustering is accounted for.

— (new) → Chapter 4, pp.133 & 135

Added a paragraph motivating the need for a Bayesian test to quantify evidence for the null followed by Bayes factor results (Table 4.7, p.135) for six monkey × magnitude-comparison combinations. Reported honestly: five of six comparisons show inconclusive-to-moderate evidence for the null, but one shows "some" evidence for H1 (a genuine difference) rather than the null. Notes this is set against the very large effect of reward magnitude seen elsewhere.

Chapter 4, p.139 (Fig 4.27) & p.141 (Fig 4.28) → Chapter 4, p.150 (Fig 4.28) & p.152 (Fig 4.29)

Verified the visual patterns in these figures against the underlying data — both real, not plotting errors. Fixed a p-value with a misplaced decimal and a mislabelled Monkey U/V caption reference. For both figures, added sentences explicitly noting the visually apparent trend while confirming the statistics remain as already reported in the text (non-significant).

— (new) → Chapter 4, pp.145–146

Added mixed-effects results for neuronal response at the win/lose reveal (Table 4.9). All relationships significant except the computer's bid for Monkey U; Monkey V's computer-bid relationship, not

significant under the naive regression, became significant — interpreted as a missed-opportunity/regret-like signal.

— (new) → Chapter 4, p.151 (Table 4.11, p.154)

Added mixed-effects results for the relationship between juice delivery and cell activity. No significant relationships for either monkey under either outcome, including Monkey V's win condition. Contrasted against the win/lose reveal result as a likely false positive removed by proper clustering correction.

— (new) → Chapter 4, p.158

Added mixed-effects results for the relationship between budget payout and cell activity (Table 4.13); fully confirms the naive regression.

— (new) → Chapter 5, pp.164–168

Fleshed out the Discussion with this thesis's own central findings, previously under-represented there: the WTP-divergence, the single-fractal behavioural effect and its (null) dopamine mirror, the binary-choice side-bias, the per-fractal split LME exception (Monkey V high), the matched-bid Bayes factor summary, and the central bid-dopamine mixed-effects confirmation. Also two corrections: previously stated as settled fact two naive-regression findings (Monkey V's juice-payout correlation; the loss-trial computer-bid null) that the mixed-effects re-analysis had since overturned.

— (new) → Chapter 5, p.169

New Section 5.3.2, "Individual differences between subjects", briefly discussing and reconciling recurring divergences between the two monkeys (the single-fractal effect, the starting-bid/liquid-driver reversal, the win/lose valence pattern), framed as expected individual variation rather than a limitation.

— (new) → Chapter 5, p.174

Added new Section 5.4.3, "What does the dopamine signal actually compute?", attempting to reconcile this thesis's utility-encoding findings with RPE/RL theory, causal learning, Berridge's incentive-salience (wanting/liking) framework, and other newer accounts. Only the causal-learning account is judged genuinely irreconcilable with the thesis's own interpretation (stated explicitly). Includes reference to DA response to 'predicted' outcomes and why this might not expect to extinguish.

Appendix A, p.185 → Appendix A, p.205

Added a footnote noting the logit/multinomial-logit choice rule used is equivalent to what is called the "softmax" function.

— (new) → Appendix B, p.217

Added a short paragraph on what dopamine computes under Berridge's incentive-salience framework. Bridges the existing wanting/liking discussion to prediction-error accounts and updating it with the last 5

years of debate.

Appendix E, p.212 → Appendix E, p.231 onwards

"Budget" substituted with B_{max} throughout, defined once at the start rather than introduced mid-way. Simplified the multi-line equation steps to remove repeated left-hand-side terms and redundant intermediate lines.

Appendix E, p.213 → Appendix E, p.233 onwards

Added full derivations of the cost-of-misbehaviour function for all three computer-bid distributions used in the task design and plotted in Figure 2.6 (uniform, normal, matched-normal).

Throughout → Throughout

All typographical and minor textual errors identified by the examiners e.g. missing/misplaced brackets, misspellings, missing words, incorrect equation/table/figure references, formatting inconsistencies have been corrected, along with further errors identified during subsequent proofreading passes.